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MIL-R-24195(SHIPS)
AMENDMENT-3
30 January 1970
SUPERSEDING 1/
Amendment - 2
15 May 1967

MILITARY SPECIFICATION

RECEIVER-COMPARATOR R-1407( )/URQ

This amendment forms a part of Military Specification MIL-R-24195(SHIPS), 28 December 1965.

Pages 1 and 2

\# 2.1: Delete reference to Military specification MIL-C-5015 and add the following Military specifications and standards:

SPECIFICATIONS

"MIL-Q-9858 - Quality Program Requirements.
"MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification.

STANDARDS

"MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
"MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
"MS 3102 - Connector, Receptacle, Electric, Box Mounting, Solder Contact, AN Type.
"MS 3106 - Connector, Plug, Electric, Straight, Solder Contact, AN Type."

Page 2

\# 2.2, listing: Delete and substitute:

"UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Illinois 60606.)"

\# 3.1.2: Delete and substitute:

"3.1.2 Reliability. A mean-time-between-failures (MTBF) of 10,000 hours shall be used as a design goal. The MTBF shall be predicted in accordance with method D of NAVSHIPS 93820. The equipment shall conform to level 4 of MIL-R-22732."

\# Add new paragraphs 3.1.3, 3.1.4, 3.1.5, 3.1.6, and 3.1.7:

"3.1.3 Modular construction. Where modular construction and integrated circuits are utilized, they shall be in accordance with MIL-E-16400.

"3.1.4 Maintainability. Maintainability shall be in accordance with MIL-E-16400.

"3.1.5 Blown fuse indicators. Blown fuse indicators shall be located on the front panel of the equipment and shall conform to MIL-F-19207.

"3.1.6 Time meters. Time meters are not required.

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 5820

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"3.1.7 Electromagnetic interference emission, susceptibility and equipment and systems self-compatibility. Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility shall be in accordance with MIL-E-16400. The equipment is designated class IC of MIL-STD-461."

Page 3

# 3.2.4: Delete "50" and substitute "60".

# 3.2.5: Add "except that temperature range shall be 0° to 60° C. (ambient operating)."

# 3.3.3, line 4: Delete "Gaussion" and substitute "Gaussian".

3.3.4.2, heading: Delete "Undersired" and substitute "Undesired".

# 3.3.4.3: Delete and substitute:

"3.3.4.3 Desensitization. Receiver gain shall not be affected by more than 3 db., and no error in phase tracking shall occur when an undesired signal is plus or minus 200 cycles, and 70 db stronger than the desired signal."

# 3.3.5, line 2: Delete "-167 dbm" and substitute "-157 dbm".

# 3.3.6: Delete and substitute:

"3.3.6 Receiver bandwidths. The following 3-db. bandwidths shall be provided:

(a) Intermediate frequency (IF) - 200 cycles ± 20 percent.
(b) 1 kc. envelope output - 200 cycles ± 20 percent.
(c) Servo predetection - 40 cycles nominal.
(d) Servo postdetection - Approximately 0.005 cycle nominal at 20 kc."

Page 5

# 3.3.7.2, second sentence: Delete "reactive" and substitute "react".

3.3.11.1, line 3: Delete "(minimum)" and substitute "(nominal)".

Page 7

# "3.3.16: Delete and substitute:

"3.3.16 Power requirements. The receiver-comparator, excluding the chart drive motor, shall operate on 115 volts, a.c. ± 10 percent, 50 to 400 cycles, single phase, or 26 ± 4 volts, d.c. negative ground. Power consumption shall not exceed 30 watts. A detachable line cord (3-wire) and plug conforming to J-C-175 (no smaller than 18 AWG conductor size) shall be supplied for connection to the 115-volt, a.c. source. Cable shall attach to the rear of the equipment through a chassis receptacle MS3102R-14S-7P and mating cable connector MS3106R-14S-7S in accordance with MS 3102 and MS 3106. Chassis connector MS3102R-12S-3P shall be located on the rear of the equipment for connection to the d.c. source. Mating cable connector MS3106R-12S-3S in accordance with MS 3106 shall be supplied. The equipment shall operate from the d.c. source in the event of a.c. power failure with no interruption of any function except the chart recorder. Chart drive motor shall operate on 115 volts, a.c., ± 10 percent, 60 cycles, single phase. Power frequencies other than 60 cycles will not require proper chart rate."

Page 8

# 3.3.19.1, line 1: Delete "is" and substitute "if".

3.3.20: First and second sentences, delete and substitute "The receiver shall be capable of operation in conjunction with an external commutator. The receiver shall be blanked upon command from the commutator."; line 7, delete "1 volt applied" and substitute "2 volts maximum".

# 4.1: Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable



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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

# Add new paragraph 4.1.1:

"4.1.1 Quality program. The supplier shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858."

# 4.2, line 2: Delete "Bureau of Ships" and substitute "Naval Ship Engineering Center."

# 4.3, line 2: Delete "bureau" and substitute "command".



# Table I: Make changes as follows:

(a) Group I, line 7: Delete "radio interference and undesired radiation" and substitute "Electromagnetic interference emission, susceptibility, and equipment and system self-compatibility".
(b) Group III, line 1, add "$C^{o}$ to $49^{o}$ C. (ambient operating)."



# 4.5.2.1, lines 2 and 3: Delete "radio interference and undesired radiation" and substitute "electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility"; delete the second "if" and substitute "it".

# 4.5.4, line 2: Delete "interference and undesired radiation and substitute "electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility".



# 4.6.5.3, line 4: Delete "31.6 mv" and substitute "3.16 mv". Add new sentence "The receiver shall faithfully track standard frequencies, with no phase error, under the condition specified herein."



4.6.9, line 11: Delete "Sevo" and substitute "Servo".



# 4.6.22: Delete and substitute:

"4.6.22 Reliability. Tests shall be performed as specified in MIL-R-22732, level 4, to determine conformance with 3.1.2. The equipment shall be operated with 19.9-kc. input at 3.1 uv level; standard response shall be obtained throughout this test. Any break in tracking shall constitute a failure."

# Section 5: Under "PREPARATION FOR DELIVERY" add the following:

"(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3)."

# 5.1.1.2, line 6: Delete "Ratings, Rules and Regulations" and substitute "Rules".



# 5.2.1.1: Delete and substitute:

"(5.2.1.1 Preservation and packaging, packing, and marking. The equipment, accessories, repair parts, and technical publications shall be preserved and packaged level A or C, packed level A or B, and marked in accordance with MIL-E-17555.)".

# 6.1, item b: Delete and substitute:

"(b) Level of preservation and packaging, packing, and marking if other than specified in 5.1 (see 5.2)."



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# 6.2 and 6.2.1: Delete and substitute:

"6.2 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24195 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.2.1).

"6.2.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120."



# Add new paragraph 6.3:

"6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped."

Preparing activity:
Navy - SH
(Project 5820-N480)



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